

Jennifer Ren

📞 US: 408 966 5023 | ✉️ jenren@berkeley.edu | [in jennifer-ren](https://www.linkedin.com/in/jennifer-ren)

SKILLS

- Languages** SystemVerilog, Verilog, Python, C, C++, Java, x86/RISC-V Assembly, LaTeX
- EDA Tools** Synopsys VCS, Synopsys Verdi, Cadence Genus, Cadence Innovus, Cadence Joules
- Other Tools** LTspice, Logisim, MATLAB, Oscilloscope, Spectrum Analyzer

PROJECTS

3-Stage Pipelined RISC-V CPU (EECS151)

2026

SYSTEMVERILOG | SYNOPSYS VCS, VERDI | CADENCE GENUS, INNOVUS, JOULES

- Designed a 3-stage pipelined RISC-V CPU in SystemVerilog
- Implemented hazard detection, forwarding, stalls, and branch/jump flushing
- Verified with assembly tests and post-synthesis simulation; ran synthesis, place-and-route, and power analysis

Procedurally Generated Dungeon Game

2025

SOFTWARE PROJECT

- Built a seed-based procedural generator for game floorplan using Java
- Implemented persistent save/load functionality and game state management

CO₂ Database Query System with Visualization

2024

PYTHON | SQL | MATLAB | BEAUTIFUL SOUP

- Scraped climate data, stored it in a structured SQL database, and built a server-client query system
- Visualized trends in MATLAB with scatter plots, histograms, and linear regression

EXPERIENCE

Machine Learning Intern

Jan 2026 – Present

BFAI SEMICONDUCTOR | PYTHON, PYTORCH, COMPUTER VISION

- Developed ML pipelines for semiconductor defect detection using classification, detection, and segmentation models
- Built preprocessing and augmentation workflows for crack and void detection in Docker-based environments

Senior Mentor, CSM16A

Jan 2025 – Present

COMPUTER SCIENCE MENTORS AT BERKELEY

- Led tutoring sessions and created supplemental notes and resources for 100+ students
- Supported students in signals, linear algebra, and data processing concepts

Project Lead, Cal Women's Network Mentorship Program

Aug 2025 – Dec 2025

BERKELEY DATA DISCOVERY

Berkeley, CA

- Cleaned and analyzed membership data using SQL, pandas, and Matplotlib
- Prototyped a mentorship platform and matching algorithm for 200+ members

Academic Intern

Jan 2025 – May 2025

CS61A COURSE STAFF

Berkeley, CA

- Taught in lab and office hours, helping students debug Python and understand core concepts
- Developed teaching strategies using extra practice problems and visual explanations

EDUCATION

B.S. in Electrical Engineering and Computer Sciences

Aug 2024 – May 2028

UNIVERSITY OF CALIFORNIA, BERKELEY | GPA: 3.75/4.00

Berkeley, CA

Coursework: Data Structures & Algorithms, Computer Architecture, Digital Design & IC, Machine Learning, Efficient Algorithms & Intractable Problems, Foundations of Signals & Systems, Circuits & Devices, Discrete Math & Probability, Linear Algebra & Differential Equations, Multivariable Calculus

Interests: Graphic Design/Painting, Competitive Figure Skating, Poker